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The Expensive Brain: A framework for explaining evolutionary changes in brain size☆

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ABSTRACT

To explain variation in relative brain size among homoiothermic vertebrates, we propose the Expensive Brain hypothesis as a unifying explanatory framework. It claims that the costs of a relatively large brain must be met by any combination of increased total energy turnover or reduced energy allocation to another expensive function such as digestion, locomotion, or production (growth and reproduction). Focusing on the energetic costs of brain enlargement, a comparative analysis of the largest mammalian sample assembled to date shows that an increase in brain size leads to larger neonates among all mammals and a longer period of immaturity among monotokous precocial species, but not among the polytokous altricial ones, who instead reduce their litter size. Relatively large brained mammals, altricial and precocial, also show reduced annual fertility rates as compared to their smaller brained relatives, but allomaternal energy inputs allow some cooperatively breeding altricial carnivores to produce even more offspring in a shorter time despite having a relatively large brain. Thus, the Expensive Brain framework explains why brain size is linked to life history pace in some, but not all mammalian lineages. This framework encompasses other hypotheses of energetic constraints on brain size variation and is also compatible with the Brain Malnutrition Risk hypothesis, but the absence of a mammal-wide correlation between brain size and immature period argues against the Needing-to-Learn explanation for slower development among large brained mammals.

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Introduction

Mammals and birds show dramatic interspecific variation in the size of their brains, both in absolute terms and after statistically removing body size correlations. Explaining variation in relative brain size is important, because the latter corresponds to intellectual or cognitive performance (Iwaniuk et al., 2001; Lefebvre et al., 2004; Sol et al., 2005; Deaner et al., 2007). Palaeoanthropologists in particular are interested in this question, because the size of the human brain is roughly three times that of our closest relatives, common and pygmy chimpanzees, who have maintained more or less similar brain sizes since the split with the last common ancestor. This spectacular departure in a rather short period of time is one of the core trends in human evolution.

Explanations for brain size variation have focused largely on benefits, such as dealing with the challenges of competition and cooperation imposed by living in social groups (e.g., Dunbar and Shultz, 2007). However, selection favours changes that produce a net benefit (i.e., also takes costs into account). To date, the work that considered the cost side of the equation has not been integrated into mainstream theory on cognitive evolution (but see Martin, 1981, 1996; Aiello and Wheeler, 1995). Consideration of costs would complement ideas positing benefits to larger brains by revealing the conditions under which such positive selection pressures would be capable of producing actual increases in brain size. Despite considerable efforts during the past decades (reviewed in Deacon, 1990), various hypotheses on brain size variation have been mostly depicted as alternatives instead of possible parts of an overarching framework, played out with different emphases in different mammalian groups. Here, we propose such an explanatory framework based on common energetic principles underlying mammalian brain size variation that yields detailed predictions for the different adaptive packages found in different taxa.

Brain tissue is energetically expensive, requiring nearly an order of magnitude more energy per unit weight than many other somatic tissues (Mink et al., 1981). Moreover, the energetic needs of the brain cannot be temporarily reduced as is possible for other organs (Karasov et al., 2004; Bauchinger et al., 2005). The proportion of metabolic energy shunted to the brain in the resting state is

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appreciable: around 13% and 20% in adult chimpanzees and humans, respectively (Mink et al., 1981; Holliday, 1986); some 8.5% in small mammals such as the mouse (*Mus musculus*); and an average for homoiothermic vertebrates of over 5% (Mink et al., 1981). It tends to be even higher for immature animals, whose brains tend to make up a higher proportion of total body weight (Holliday, 1986) and are still differentiating even if fully grown. The high proportion of energy necessarily allocated to brain tissue may therefore impose serious constraints on the evolution of brain size, despite benefits to increased brain size.

The aim of this paper is to test the value of the Expensive Brain approach to explain brain size evolution in mammals in general. Focusing on the size of the brain as a whole rather than on specific regions is warranted from an energetic perspective. In addition, an individual's cognitive performance is linked to the size of its whole brain, corrected for the size of its body, both between (Jerison, 1973; Deaner et al., 2007) and even within species such as humans (McDaniel, 2005). One reason for this may be that the composition of brains across species within a lineage shows obvious regularities that suggest that changes in one region are necessarily linked to a fairly large extent to changes in others (Finlay and Darlington, 1995; Jerison, 2001).

The costs of increased brain size can be met in two distinct but complementary ways, illustrated in Figure 1: (1) by increasing energy intake or (2) by changing energy allocations. Increased energy intake is the more risky strategy because brain starvation needs to be avoided (through a combination of improved continuity of intake and fat storage), but there is evidence for it. Thus, in support of the direct metabolic constraints hypothesis (Armstrong, 1983; Hofman, 1983), we found a significantly positive relationship between relative brain mass and relative basal metabolic rate (BMR, post absorptive metabolic rate of nonreproductive adults during rest and in a thermo neutral state) in mammals, which explains up to 15% of brain mass variation in precocial mammals and even 23% in primates (independent contrasts analyses, Isler and van Schaik, 2006b; Isler et al., 2008).

Modified energy allocation is the second plausible option. Allocation targets in animals can be divided into maintenance and production (which comprises growth and reproduction). Tradeoffs within the maintenance category are likely. The Expensive Tissue hypothesis (Aiello and Wheeler, 1995) examines tradeoffs among energy-hungry organs. Given the size of the organs involved, compensatory increases in brain size accompanying evolutionary reductions in gut size should be the most profitable among these tradeoffs, and is indeed well supported for anthropoid primates (Aiello and Wheeler, 1995; Fish and Lockwood, 2003), although Aiello et al. (2001) show that data selection and analytical technique substantially influence the results obtained. Also, the Expensive Tissue hypothesis was not confirmed for bats (Jones and MacLarnon, 2004) or birds (Isler and van Schaik, 2006a), perhaps because these latter taxa have small gastro-intestinal tracts. This mixed support has created uncertainty about the status of the Expensive Tissue hypothesis.

A more general hypothesis for tradeoffs among different maintenance functions is supported for birds, where a negative correlation is found between the relative size of pectoral muscles and the brain (Isler and van Schaik, 2006a). Muscle tissue is not as costly per unit weight at rest as brain tissue, but the large size of some muscles makes them sizeable allocation targets (skeletal muscles use 20% of resting metabolism in humans and 30% in rats; Rolfe and Brown, 1997). This suggests that relatively under muscled organisms, such as primates, may be able to afford larger brains in part through this mechanism. Accordingly, we proposed that an energetic tradeoff between the brain and skeletal muscles might not only act during rest, but also during locomotion. Although the amount of skeletal muscle mass does not differ between modern humans and apes, a chimpanzee spends much more of its daily energy budget on locomotion than a similarly sized human hunter-gatherer (Pontzer and Wrangham, 2004; Isler and van Schaik, 2006a). Thus, a tradeoff between the energy used for locomotion and for the brain may have played an important role in human evolution, when in early *Homo* the energetically less efficient, australopithecine-like form of bipedalism evolved into a modern striding gait, probably because climbing was abandoned and anatomical compromises were no longer needed.

The other major allocation tradeoff is with production. Large brained organisms may be able to reduce the energy allocated to production (i.e., growth and reproduction) if the survival benefits of increased brain size outweigh the inevitable reduction of the annual fertility rate. Placental mammals follow two fundamental strategies for reproduction (Portmann, 1939): altricial species produce a litter of multiple blind, relatively small, helpless neonates, whereas precocial species give birth to well-developed, relatively large, often single young. In altricial, polytokous (several offspring per reproductive event) mammals, we predict a tradeoff between the number of offspring (litter size) and brain size, because litter size is easily adjusted to environmental conditions or the vigour of the individual mother. On the other hand, precocial, monotokous (single offspring per reproductive event) mammals cannot reduce litter size, and thus we expect relatively large brained species to distribute increased reproductive costs over a longer time period. We predict longer interbirth intervals and prolonged gestation and lactation lengths in large brained species. In both groups, reduced fertility rates should be compensated for by an increased reproductive lifespan to maintain demographic viability.

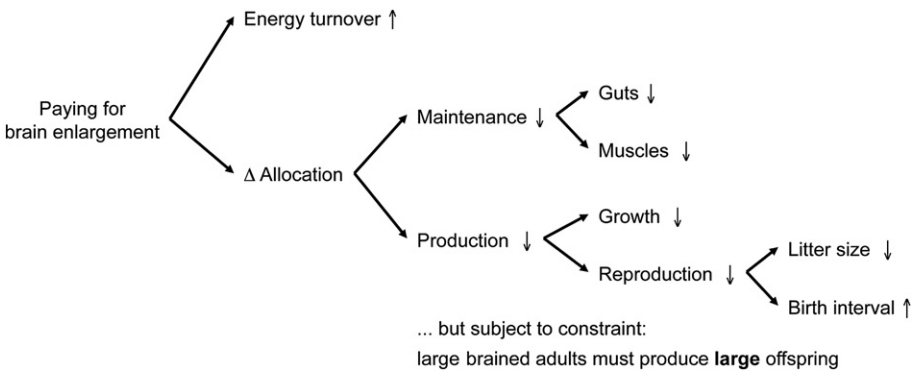


Fig. 1. Diagram of the Expensive Brain framework. Alternative ways of paying for the increased energy demand of a relatively large brain are not mutually exclusive, but may be expressed to a different degree in various taxonomic groups.

In cooperatively breeding species, the mother receives energy subsidies during the strenuous reproductive period, or the offspring receives high quality food by the father or other group members. In these species, we therefore expect the tradeoff between brain size and fertility to be weakened or absent. In order to test this prediction, we compare terrestrial carnivore families with common allomaternal help (Canidae, Herpestidae, and Hyaenidae), with those families that do not usually have helpers (Felidae, Viverridae, Mustelidae, Procyonidae, Ursidae) and should be subject to the same tradeoffs as in other altricial mammals.

Other effects of increased brain size on reproduction are expected as well. Relatively large brained species produce large brained, and therefore energetically more expensive, neonates relative to neonate body mass (Figure 2). More importantly, the Brain Malnutrition Risk hypothesis (Deaner et al., 2003) argues that such a large brained offspring should be more vulnerable to unexpected shortages of energy supply due to the need for continuity of the energy supply to the brain, and would, *ceteris paribus*, lead to increased infant mortality compared to their smaller-brained relatives. In response, relatively large brained species may therefore have to invest even more energy into reproduction and produce larger neonates. No tests of this prediction have been conducted to date.

In this paper, we report a series of multivariate tests of the predictions for production (i.e., life history characteristics) flowing from the Expensive Brain framework (Table 1). These tests are conducted on the largest sample of mammalian life history and brain data assembled to date. We will then discuss the relation of the Expensive Brain framework to several earlier hypotheses concerning the possible role of energetics in brain size evolution.

Table 1
Predictions on correlations between life history variables and brain size flowing from the Expensive Brain framework

Relative to small brained monotokous precocial mammals, large brained ones exhibit (controlling for body size):
longer development periods (gestation, lactation, and total immature periods)
reduced annual fertility by prolonged interbirth intervals (fewer litters per year)
increased adult survival, and thus ultimately prolonged maximum lifespan
larger neonates
Relative to small brained altricial mammals, large brained ones exhibit (controlling for body size):
reduced annual fertility by fewer offspring per litter
similar development periods
increased adult survival, and thus ultimately prolonged maximum lifespan
larger neonates

Materials and methods

We compiled a broad data set on average brain and body mass, as well as life history parameters of eutherian mammals (1247 species, excluding *Homo sapiens*). Life history and brain and body mass data were taken from published compilations (sources are listed in the Supplementary Online Materials [SOM]: supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhevol.2009.04.009). If data from several populations or subspecies were available, the median value was used as a species mean. The following life history variables were compiled: gestation length (excluding periods of delayed implantation), lactation length (birth to weaning age), sexual maturity, maximum lifespan, and annual fertility of an average female as the product of the number of offspring per litter times the number of litters per year. Reaching sexual maturity was defined as the age at which first conception took place (age at first reproduction minus gestation length) and was taken for females only, if available. Maximum lifespan represents the possible longevity under optimum conditions, and we therefore did not discern between wild and captive records.

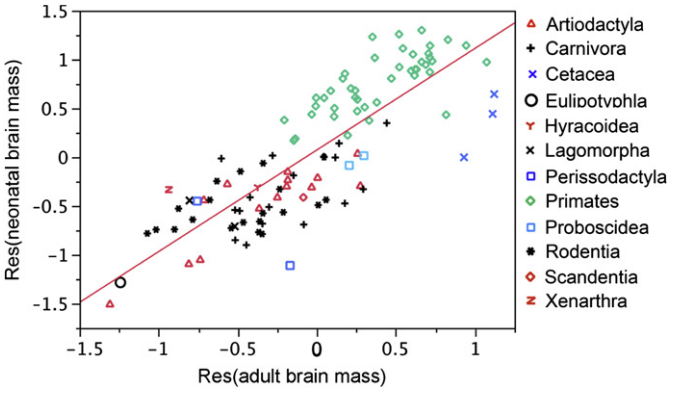


Fig. 2. Least-squares regression of residuals of neonate brain mass (vs. neonate body mass) versus residuals of adult brain mass (vs. adult body mass). In both precocial and altricial mammals, relatively large brained species produce relatively large brained neonates (all mammals: $n = 108$, $r^2 = 0.685$, $p < 0.0001$, IC $p < 0.0001$; altricial polytokous species: $n = 28$, $r^2 = 0.223$, $p = 0.011$, IC n.s.; precocial monotokous species: $n = 61$, $r^2 = 0.675$, $p < 0.0001$, IC $p < 0.0001$; primates: $n = 44$, $r^2 = 0.191$, $p = 0.003$, IC $p = 0.012$).

As data quality is an important issue in comparative studies, we carefully checked original sources of brain size data to include only adult individuals. For species with sexual dimorphism in body size, female values were preferred if available (details in the SOM). For approximately two thirds of the species, brain size is represented by endocranial volumes (mainly carnivores, rodents, and primates) and for one third by the mass of fresh brains (mainly bats, cetartiodactyls, and smaller orders). As the type of data is quite homogeneous within orders, and brain mass is given for all special orders such as Xenarthra, Proboscidea, Sirenia, or Hyracoidea, we are confident that a possible allometric relationship between brain mass and endocranial volume, as found in domesticated mammals by Röhrs and Ebinger (2001), does not seriously affect our analyses. At present, systematic bias seems to be small relative to error variances (cf. Isler et al., 2008 for primates) and it cannot be responsible for any patterns detected in multiple regressions, especially if the same type of data is available for closely related species (as independent contrasts analyses are particularly vulnerable to error effects near the tips of the trees; cf. Purvis and Webster, 1999). Neonate brain mass data have been taken from the compilation of Sacher and Staffeldt (1974).

The proposed tradeoffs assume equal total energy budgets, but these are known from only very few mammalian species and vary dramatically according to environmental conditions. In mammals, basal metabolic rate (BMR) is a good indicator of average daily energy expenditure or field metabolic rate (White and Seymour, 2004). Thus, we use BMR as a covariate to test whether observed correlations are in fact based on differences in metabolic rates. Data on mammalian BMR and the concomitant body mass have been compiled using White and Seymour (2003) and Lovegrove (2000, 2003). Basal metabolic rate is defined as the rate of metabolism of adult, postabsorptive, nonreproductive individuals during the resting phase in their thermoneutral zone. For primate BMR data, we checked the original sources, because we found that some of the results compiled in White and Seymour (2003) and Lovegrove (2000) contain data of immature, active, or anaesthetised animals. For a complete list of primate BMR data see Isler et al. (2008).

Taxonomy follows Groves (2005) for primates and Nowak (1999) for all other mammals. Species were defined as precocial if the young open their eyes at birth or shortly thereafter and as altricial otherwise. The data set was divided into the following two groups: precocial, monotokous mammals (average number of offspring less

than 1.5 per litter) and altricial, polytokous mammals. Most families of Chiroptera produce one single, large offspring after a long gestation, but this opens its eyes only after some days. Thus, all Chiroptera are omitted in the analyses. In total, we included 114 precocial monotokous genera from 10 orders, derived from data of 161 species and 137 altricial polytokous genera from 7 orders, derived from data of 209 species. In analyses including maximum lifespan, species from the order Cetacea were excluded due to the high unreliability of lifespan data of these long-lived, aquatic animals.

Terrestrial carnivores were divided into those families with help during breeding (Canidae, Herpestidae, Hyaenidae) and those with maternal care only (Felidae, Mustelidae, Viverridae, Procyonidae, Ursidae). To obtain reasonable sample sizes, the two carnivore subgroups were analysed at the species level while including only those species for which all studied life history parameters were available.

All comparative analyses of brain size variation require the removal of the effects of body mass in order to test the hypotheses adequately. We used multiple least-squares regressions with each life history variable as the dependent variable, and brain and body mass as independent variables (Garcia-Berthou, 2001), but for illustration purposes we show the relationship between residuals of life history and the residuals of brain mass, both calculated from least-squares regressions versus body mass. However, body mass of the individuals for which BMR was measured often differs considerably from the species' mean body mass. Thus, we calculated a corrected BMR matching the species mean body mass by using an overall slope of 0.7236 from the least-squares regression of BMR versus body mass ($n = 382$ mammals).

Phylogenetic relationships are those proposed by Bininda-Emonds et al. (2007). In large comparative analyses, the problem of different degrees of relatedness between species must always be addressed. The method of calculating phylogenetically independent contrasts, as proposed by Felsenstein (1985), has been widely used to reduce the impact of phylogenetic influences on comparative analyses (Harvey and Pagel, 1991; Pagel, 1992). Contrasts were calculated using the CRUNCH algorithm in CAIC (Purvis and Rambaut, 1995). Molecular branch length estimations were taken from the species-level supertree of Bininda-Emonds et al. (2007), and their appropriateness tested with the program Continuous (Pagel, 1992) and the regressions of absolute contrasts versus their standard deviations (as recommended by Garland et al., 1992). Details are listed in the SOM. According to these tests, we used equal branch lengths to standardise contrasts for all data sets of altricial polytokous mammals and for carnivores, and estimated branch lengths based on the molecular clock for data sets of precocial monotokous mammals. Maximum likelihood estimates of the parameter Lambda were close to 1 for all data sets, indicating that the assumption underlying contrasts that evolution proceeded according to a 'random walk' is justified and the method of independent contrasts can be applied (Pagel, 1992).

The independent contrasts technique magnifies the effects of measurement error (see e.g. Ricklefs and Starck, 1996; Martin et al., 2005), especially if there are many closely related species of similar body size in a genus and there is therefore no guarantee to obtain the most reliable results for a particular data set (Rohlf, 2006). Thus, we always report results of analyses based on log-transformed species data (TIP, denoted here as raw), as well as from independent contrasts analyses (PIC, denoted here as IC). To minimise the influence of error variation that leads to discrepancies between the methods, we reduced the number of species included in the analyses to those for which all the following life history parameters were available: gestation length, length of the lactation period, age at first conception (sexual maturity), maximum recorded lifespan (in the wild or in captivity), litter size (average number of offspring per litter), average number of litters per year, and neonate mass. If all parameters for several congeneric species were available, we calculated a genus average. In addition, we ran species-level analyses for all available data and did not find differences in the level of significance of the results. Due to the more limited sample, all analyses including BMR as a covariate were done at the species level.

If results differ between raw and independent contrasts analyses, we adhere to the following rationale for interpretation (see also Barrickman et al., 2008): if a grade shift is detected that influences raw data regressions, IC results are given preference. Otherwise, if IC results do not confirm significant relationships found in raw data analysis, we tested whether omission of those contrasts with absolute studentised residuals larger than three yield more consistent results (Jones and Purvis, 1997). However, this procedure did not lead to any changes in the level of significance and results are thus not shown.

Results

In all mammals, regardless of developmental mode at birth, we predicted that neonates should become larger to compensate for the increased impact of starvation on survival in large brained animals (note that we always control for the effect of body mass [e.g., "large brained" does always mean "relatively large brained," etc.]). The results confirmed this prediction in both monotokous precocial species and polytokous altricial species (Table 2), and remained significant when phylogenetic nonindependence was controlled for by contrast analysis (Figure 3). Thus, large brained species must invest more in each offspring they produce.

In precocial, monotokous lineages, we predicted a slowdown in development among large brained taxa if body mass is controlled for, which should be compensated for by an increase in adult lifespan. Table 2 shows that in our sample of monotokous precocial mammals, this was indeed the case: gestation, lactation, and the total immature period, as well as maximum lifespan, increased with brain size, whereas annual fertility (offspring produced per year) and litters produced per year decreased.

In altricial lineages, we predicted a reduction in litter size among large brained taxa, if body mass is controlled for, which should be accompanied by increased life span in compensation. This was confirmed by our results for the altricial polytokous mammals (Table 2): brain mass was positively correlated with lifespan and negatively correlated with annual fertility rates and litter size. But, unlike in precocial mammals, gestation length and the total length of the immature period was not correlated with brain mass (Figure 3). In this group, reduced fertility was achieved by reducing litter size rather than by reducing the number of reproductive events per year.

Furthermore, we found that carnivore families in which mothers often experience help from either the male or other group members in rearing the offspring are characterised by a complete absence or even a reversal of brain-mass-life history correlations (Table 2). In raw species-level analyses, cooperatively breeding large brained species even managed to produce larger litters in a shorter time period than smaller brained species, whereas in IC analyses, only lactation length was significantly negatively correlated to brain mass. Carnivores without helpers showed the pattern expected for altricial, polytokous mammals.

The question arises whether the strategies of raising BMR, reducing the number of offspring per litter, and lengthening development periods can all be used simultaneously by large brained species, or whether they represent alternative strategies. To test this, we conducted a multiple regression with brain mass as the response variable and body mass, BMR, litter size, and length of either gestation or lactation as independent effects (Table 3).

Table 2
The main results of multiple regressions with each life history parameter as dependent variable, and ln brain mass and ln body mass as independent variables[a]

	n	Gestation length		Lactation length		Sexual maturity	
	Raw/IC	Raw	IC	Raw	IC	Raw	IC
Precocial, monotokous mammals	114/101	**< 0.001 (+)**	**0.002 (+)**	**< 0.001 (+)**	**0.001 (+)**	**< 0.001 (+)**	**0.008 (+)**
Altricial, polytokous mammals	137/101	0.636 (+)	0.074 (+)	**0.002(+)**	**0.024 (+)**	0.365 (+)	0.858 (+)
Carnivores with helpers	23/17	**< 0.001 (−)**	0.100 (−)	**0.001 (−)**	**0.001 (−)**	**0.003 (−)**	0.067 (−)
Carnivores without helpers	54/45	0.866 (+)	0.173 (+)	0.625 (−)	0.627 (−)	0.363 (+)	0.528 (+)
	n	Neonate mass		Litter mass		Maximum lifespan[b]	
	Raw/IC	Raw	IC	Raw	IC	Raw	IC
Precocial, monotokous mammals	114/101	**< 0.001 (+)**	**< 0.001 (+)**	**< 0.001 (+)**	**< 0.001 (+)**	**< 0.001 (+)**	**0.049 (+)**
Altricial, polytokous mammals	137/101	**0.008 (+)**	**0.019 (+)**	0.532 (+)	0.412 (+)	**0.003 (+)**	**0.050 (+)**
Carnivores with helpers	23/17	0.274 (−)	0.939 (−)	**0.039 (+)**	0.380 (+)	**0.024 (−)**	0.705 (−)
Carnivores without helpers	54/45	0.125 (+)	**0.006 (+)**	0.492 (+)	0.184 (+)	**0.007 (+)**	**< 0.001 (+)**
	n	Annual fertility		Litter size[c]		Litters per year	
	Raw/IC	Raw	IC	Raw	IC	Raw	IC
Precocial, monotokous mammals	114/101	**< 0.001 (−)**	**0.002 (−)**	0.061 (−)	0.885 (−)	**< 0.001 (−)**	**0.001 (−)**
Altricial, polytokous mammals	137/101	**0.026 (−)**	**0.019 (−)**	**0.014 (−)**	**< 0.001 (−)**	0.375 (+)	0.976 (−)
Carnivores with helpers	23/17	0.063 (+)	0.843 (−)	**0.013 (+)**	0.374 (+)	0.544 (−)	**0.018 (−)**
Carnivores without helpers	54/45	**0.020 (−)**	0.068 (−)	0.104 (−)	**0.038 (−)**	0.177 (−)	0.706 (−)

[a] The table shows the significance of the regression coefficients of brain mass on the life history variables, controlling for body mass, followed by the direction of the correlation in parentheses (significant values in bold). The table provides an overview of results, whereas full statistical details including p-values of the effects of body mass and r^2 values are given in the SOM.

[b] In the analysis of maximum lifespan, cetaceans were excluded from the group of precocial monotokous mammals due to unreliable data (thus, $n = 99$ instead of $n = 114$ species and $n = 86$ rather than $n = 101$ contrasts).

[c] Note that litter size is also analysed in monotokous mammals because some of them occasionally produce twins.

Controlling for the impact of body mass, BMR, and litter size, large brained precocial monotokous mammals exhibited a longer gestation and lactation period. For altricial species, only lactation length is positively correlated to brain mass, but not gestation length (again controlling for body mass, BMR, and litter size). In other words, the partial correlation between developmental periods and BMR would be significantly negative if body mass, brain mass, and litter size were held constant. Overall, this shows that raising BMR and prolonging developmental periods are alternative strategies that are not necessarily pursued simultaneously.

There is another possible source of variation in maternal energy shunted to the offspring that has not been included so far, because adequate data are largely missing: a rise in maternal metabolic rate during gestation that is not proportional to BMR. Large brained mothers might be able to disproportionately increase their gestational metabolic rate during gestation. Thus, we tested whether the large body mass of neonates of large brained species is completely determined by the combined effect of the energetic benefits of increased BMR, longer gestation, and reduced litter size, or whether brain mass is still a significant effect if the other parameters are controlled for (Table 4). Controlling for body mass, BMR, gestation length, and litter size, we found there is still a positive correlation between brain mass and neonate mass in altricial species, but not in precocial species (negative correlation in raw data analysis, no significant correlation in IC analysis). In other words, in the same period of time, controlling for BMR and litter size, a large brained altricial mother produces individually heavier neonates than a small brained mother. To invest this additional energy, large brained altricial, polytokous mammals might be able to disproportionately increase their metabolic rate during gestation.

Discussion

The Expensive Brain hypothesis evaluated

The Expensive Brain hypothesis argues that evolutionary increases in brain size can be paid for by any combination of increased energy turnover and reduced allocation to other targets, such as maintenance and production, and therefore often reproduction. The analyses of a large sample of mammalian species showed that relatively large brained mammals reduce their fertility rate by either reducing the number of offspring per litter in multiple litters or by distributing the reproductive effort over a longer time period if they have single births. In either case, the reduced fertility is compensated by a longer lifespan, although not generally enough to maintain net reproductive rates at the same level (Isler and van Schaik, 2009). However, any form of cooperative breeding, producing energy inputs into the female or the dependent infants by the male or other group members, allows some large brained carnivores to even increase their reproductive rate, whereas maximum lifespan is not prolonged. Thus, the main predictions of the Expensive Brain hypothesis are fully supported by the results of the comparative analyses.

The fundamental mammalian dichotomy between altricial and precocial development modes provides an opportunity for understanding brain size variation and its energetic basis by testing the predictions flowing from our framework in more detail. Figure 4 summarises our findings. For all mammals, large brained species must produce larger neonates, exacerbating the problem that increases in brain size must be paid for by reductions in allocation to production or compensatory increases in metabolic turnover. Large brained altricial mammals produce smaller litters containing individually larger neonates than relatively small brained altricial mammals, but energy subsidies from the outside, as in cooperative breeders, destroy this relationship. Because the number of offspring per litter can easily be adjusted to energetic constraints using existing physiological mechanisms, gestation length is not necessarily longer for large brained altricial mammals. Most notably, the Expensive Brain framework yields a convincing explanation of why the relationships between brain size and the length of immature periods differ completely between the two major mammalian groups that differ in development mode, precocial and altricial species.

Further predictions of the Expensive Brain hypothesis were tested in multivariate analyses including BMR as a covariate. We found that raising BMR and prolonging immature periods are alternative strategies that may or may not be pursued simultaneously by relatively

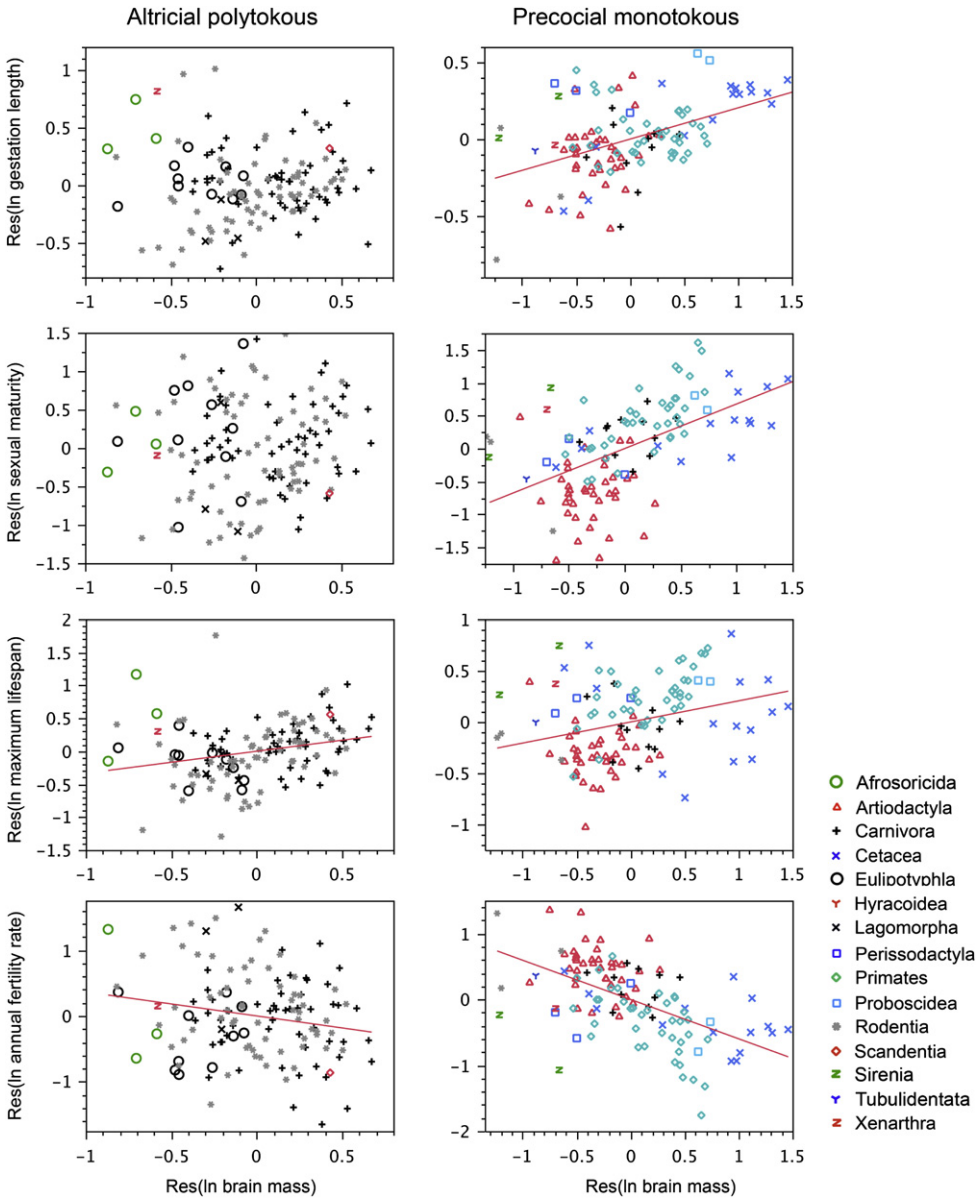


Fig. 3. Brain mass and the length of developmental periods are positively correlated in precocial monotokous mammals, but not in altricial polytokous mammals. In both groups, maximum lifespan and brain mass are positively correlated and annual fertility rates and brain mass are negatively correlated, controlling for body mass. For statistics and sample sizes, see Table 2.

Table 3
Multiple regressions with ln brain mass as dependent variable, and ln body mass, ln BMR, ln litter size, and ln gestation as independent variables (a), or all these with ln lactation length instead of gestation (b)[a]

a)	*n*	Gestation length		Litter size		BMR		Body mass	
	Raw/IC	Raw	IC	Raw	IC	Raw	IC	Raw	IC
Precocial, monotokous mammals	43/40	**< 0.001 (+)**	**0.006 (+)**	0.883 (−)	0.647 (−)	**< 0.001 (+)**	0.059 (+)	**0.379 (+)**	**0.004 (+)**
Altricial, polytokous mammals	147/116	0.345 (+)	0.136 (+)	**0.031 (−)**	0.079 (−)	**< 0.001 (+)**	**0.032 (+)**	**< 0.001 (+)**	**< 0.001 (+)**
b)	*n*	Lactation length		Litter size		BMR		Body mass	
	Raw/IC	Raw	IC	Raw	IC	Raw	IC	Raw	IC
Precocial, monotokous mammals	43/40	**< 0.001 (+)**	**0.008 (+)**	0.850 (+)	0.587 (−)	**< 0.001 (+)**	**0.023 (+)**	**0.095 (+)**	**0.005 (+)**
Altricial, polytokous mammals	147/116	**< 0.001 (+)**	**0.002 (+)**	0.053 (−)	**0.049 (−)**	**< 0.001 (+)**	**0.014 (+)**	**< 0.001 (+)**	**< 0.001 (+)**

[a] P-values of the regression coefficients are shown (significant values in bold), followed by the direction of the correlation in parentheses.

Table 4
Multiple regressions with ln neonate mass as dependent variable, and ln body mass, ln BMR, ln gestation length, ln litter size, and ln brain mass as independent variables[a]

	N	Brain mass		Gestation length		BMR		Litter size		Body mass	
	Raw/IC	Raw	IC	Raw	IC	Raw	IC	Raw	IC	Raw	IC
Precocial, monotokous mammals	43/40	**0.049 (−)**	0.522 (+)	**0.045 (+)**	0.686 (+)	**< 0.001 (+)**	**0.012 (+)**	**0.032 (–)**	**0.003 (–)**	**< 0.001 (+)**	**< 0.001 (+)**
Altricial, polytokous mammals	147/116	**0.048 (+)**	**0.005 (+)**	**0.001 (+)**	0.241 (+)	0.994 (−)	0.704 (+)	**< 0.001 (−)**	**0.002 (−)**	**< 0.001 (+)**	**< 0.001 (+)**

[a] P-values of the regression coefficients are shown, followed by the direction of the correlation in parentheses (significant values in bold).

large brained species. Thus, if BMR is not elevated, larger brain size must be accompanied by an even longer gestation and lactation period and an increase in the total duration of immaturity. The contrast between humans and extant great apes, accompanied by a large difference in brain size, is only partly consistent with this general mammalian pattern, probably because of the behavioural innovation of allomaternal inputs.

In relatively large brained altricial mammals, the individual neonates are still heavier than in small brained species even when body mass, gestation length, basal metabolic rate, and litter size are controlled for. Thus, our findings are consistent with the idea that large brained altricial mothers might be able to disproportionately increase their field metabolic rate during gestation, whereas precocial mothers might not have this possibility (cf. Künkele et al., 2005). However, this issue remains to be studied in more detail using currently unavailable gestational metabolic rates.

Methodological considerations

To reduce error variation, we have analysed a reduced data set of species for which all analysed life history parameters are available, which indicates that a species is well studied both in captivity and in the wild. Additionally, we have used genus averages for those genera in which more than one species fulfilled the first criterion to reduce possible error variation between closely related species of similar body mass. With this reduced data set, we obtained maximum congruency between the results of raw data analyses and IC analyses. However, alternative analyses on the species level for all available data yielded very similar results. Discrepancies between results of raw data analyses and of independent contrasts are mainly found in analyses of the relatively small data set of carnivore species with allomaternal help ($n = 23$ species, $n = 17$ contrasts). They are probably due to a grade shift in relative brain size between Canidae and the other families (Herpestidae and Hyaenidae), but the small sample prevents a further splitting of the group to test this. Thus, we regard IC results to be more conclusive for this group. Likewise, some discrepancies between raw data and IC results are observed in multivariate analyses that include several independent variables (cf. Table 4 for precocials). Such analyses are notoriously sensitive to collinearity problems (e.g., Darlington, 1998), and grade shifts are difficult to detect. However, as grade shifts and interactions between life history variables and body mass are of less concern in IC analyses, again we opt for putting faith in them rather than results of raw data analyses, especially after influential contrasts with absolute studentised residuals larger than three have been omitted.

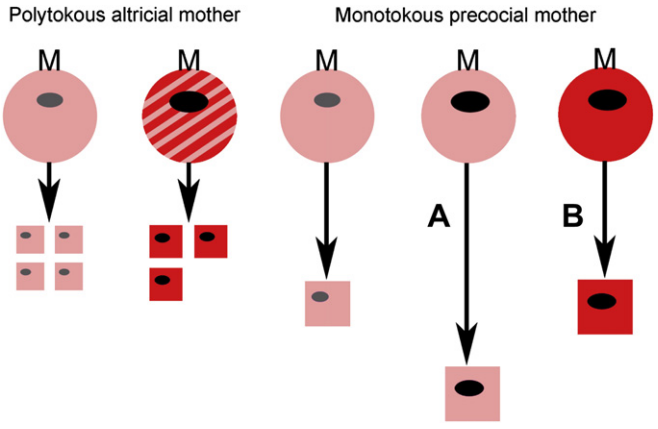


Fig. 4. How do large brained species produce large brained offspring? The lengths of the arrows represent the length of gestation, variation in offspring size is represented by the size of the box at the end of the arrow, and variation in brain size is represented by the size of the blob inside the mother's body. Relatively small brained species often exhibit a relatively low basal metabolic rate (BMR, depicted in pink), whereas relatively large brained species may exhibit an increased BMR (depicted in red). However, variation in BMR does not account completely for brain size variation in the mother or in the offspring. Large brained altricial mothers produce a smaller litter of individually heavier offspring in the same time, with or without raising BMR. Large brained monotokous precocial mothers produce a single larger offspring after a longer gestation period (A) or by increasing BMR (B) or both. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The explanatory power of the Expensive Brain hypothesis

The results indicate that the Expensive Brain framework accounts for the well known correlation between brain size and life history pace in mammals, but unlike earlier ideas (reviewed in Deaner et al., 2003; Ross, 2004) can also account for the differences in this correlation between precocial and altricial mammals. The Cognitive Buffer hypothesis (Allman et al., 1993) argues that the cognitive benefits of larger brains provide higher adult survival and thus, predicts a positive correlation between brain size and lifespan in all vertebrates. However, we found here that in social carnivores, increased brain mass per se does not necessarily entail a longer lifespan, probably because the cognitive benefits are translated into reproductive improvements relative to the expected reduction due to larger brain size. While there is no doubt that larger brains provide cognitive benefits (e.g., Sol et al., 2005, 2007; Deaner et al., 2007), the present findings suggest that these benefits do not necessarily flow into increased adult survival, but may sometimes influence reproduction more.

The Maturational Constraints hypothesis (also known as the Needing-to-learn hypothesis) claims that brain differentiation takes time, especially in order to learn vital skills (Ross and Jones, 1999; Ross, 2004). Our finding that altricial mammals with larger brains can develop as fast as those with smaller brains, and possibly develop even faster if they are provisioned by the father or other group members, suggests that maturation time is constrained by the energetic demands of brain growth and development, perhaps in concert with those of somatic growth, rather than by differentiation time linked to the amount of learning required to function as an adult.

We found that relatively large brained mammals invest relatively more energy into producing each single offspring than do smaller brained species. Following the brain malnutrition risk hypothesis (Deaner et al., 2003), we argue that a relatively large brained infant is, after birth and well into adolescence, highly vulnerable to temporal dips in the energy supply and that a relatively large neonate body mass might be needed to buffer this risk.

This is consistent with the well known fact that human neonates are relatively large and have high fat content (Kuzawa, 1998). This is also consistent with the finding that development of offspring in cooperative breeders is not delayed as their brain size increases, since the helpers buffer the food supply to the developing young.

Furthermore, the Expensive Brain framework encompasses two earlier ideas about a possible link between energetics and brain size. Based on the similarity of allometric slopes with body mass, Armstrong (1983) and Hofman (1983) were the first to suggest a direct metabolic constraint on brain size. The energetic viewpoint was largely abandoned for many years (but see Aiello and Wheeler, 1995, 1996; Aiello et al., 2001) due to the rejection of the existence of a correlation between brain mass and BMR (McNab and Eisenberg, 1989). However, this rejection was unjustified, partly due to a statistical error (Martin, 1998) and partly due to an error in rodent brain mass data (see Isler and van Schaik, 2006b). Recent analyses confirmed the significant correlation between BMR and adult brain mass, taking body mass and phylogenetic influences into account (Isler and van Schaik, 2006b). Although only part of the variation in brain size can be attributed to variation in BMR, energy turnover rates at rest must not be neglected in any attempt to understand brain size evolution. The Maternal Energy hypothesis (Martin, 1981, 1996; Martin et al., 2005) proposes a link between neonate brain mass and maternal BMR, modified by the length of gestation and possibly also lactation, because the amount of energy provided by the mother is thought to act as a constraint on the brain size of the offspring. This hypothesis was both refuted (Pagel and Harvey, 1988; Barton, 1999) and corroborated (Martin, 1998; Isler et al., 2008), the conflicting results being mostly due to methodological details. To adequately test the Maternal Energy hypothesis, we ought to include actual neonate brain mass instead of taking maternal brain mass as a proxy, which at present unduly limits sample size, especially in combination with the few available BMR measurements for large bodied primates. In any case, this hypothesis fits well into the Expensive Brain framework, which argues that selection can adjust both the length of gestation or lactation and BMR to produce neonates of the required brain and body size.

Conclusions

Using a large compilation of brain size, body mass, and life history data for 1247 placental mammalian species, we confirm an important aspect of the Expensive Brain framework: an energetically costly increase in brain size has to be met by either increasing the total energy budget of a species or by compensating changes of energy allocation to other maintenance functions, such as digestion or production (i.e., growth [in immatures] and offspring production [in adults], or a combination of these). Large brained altricial, polytokous mammals reduce their cost of reproduction by producing smaller litters, but the individual neonates are nevertheless relatively large. Reduced fertility is compensated for by a prolonged lifespan. The same is true for large brained precocial, monotokous mammals, but they cannot reduce litter size and are thus forced to prolong development periods. From the diverging strategies of the two groups of placental mammals with clearly separate development modes, we suggest that changes in brain size are indeed a cause of prolonged life history pace, but only in monotokous, precocial mammals. The development and maturation of a relatively large brain does not need more time, if sufficient energy is available (although we cannot exclude the possibility that human brains might be special in this respect). The Expensive Brain framework could therefore be a useful tool in interpreting patterns in hominin evolution (see also Isler and van Schaik, 2009).

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Supplementary data

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhevol.2009.04.009.

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